

1. Feature Name

Secure User Authorization for Shuttle Access

2. Feature Owner

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3. Feature Description

Secure User Authorization for Shuttle Access is a feature designed to provide users with a secure and personalized boarding experience when accessing an autonomous shuttle.

When a user subscribes to the **PaxAuto's** autonomous shuttle service, they are assigned a **unique code**, which is personalized and unique to each user and is used later to authorize access to the shuttle.

Once a user has booked a shuttle through the smartwatch application, the shuttle autonomously navigates to the pickup location. When the shuttle arrives and enters the boarding state, the authorization feature becomes available to the user.

From the user's perspective, accessing the shuttle is simple and intuitive. The user can authenticate their identity using one of two supported methods:

- a) By presenting a **QR code** displayed in the smartwatch application to the shuttle's RealSense camera, or
- b) By manually entering a **unique code** using the onboard keyboard.

Upon successful authorization, the user is granted access to the shuttle and can proceed to board. The door-opening action is currently represented virtually within the system and is indicated by a successful access confirmation displayed in the terminal, reflecting the shuttle's readiness for boarding.

This feature ensures that only authorized users can access the shuttle, while maintaining a user-friendly boarding process.

4. Feature Functionality/Properties

The **Secure User Authorization for Shuttle Access** feature provides a reliable and user-friendly mechanism to control access to the autonomous shuttle during boarding.

- The feature becomes available only when the shuttle arrives at the pickup location and enters the boarding state.
- Each subscribed user is assigned a unique code, which is personal and used to verify identity before boarding.
- The feature supports two alternative authorization methods, allowing flexibility for the user:
 - a) QR code-based authorization via the smartwatch application
 - b) Manual authorization via unique code entry using a keyboard
- The user may choose either authorization method without affecting the access outcome.
- Successful authorization results in access being granted to the user, allowing them to board the shuttle.
- The shuttle door opening is currently implemented as a virtual action and is indicated by a successful access confirmation shown in the system terminal.
- If authorization is unsuccessful, access to the shuttle is denied and the user is not allowed to board.

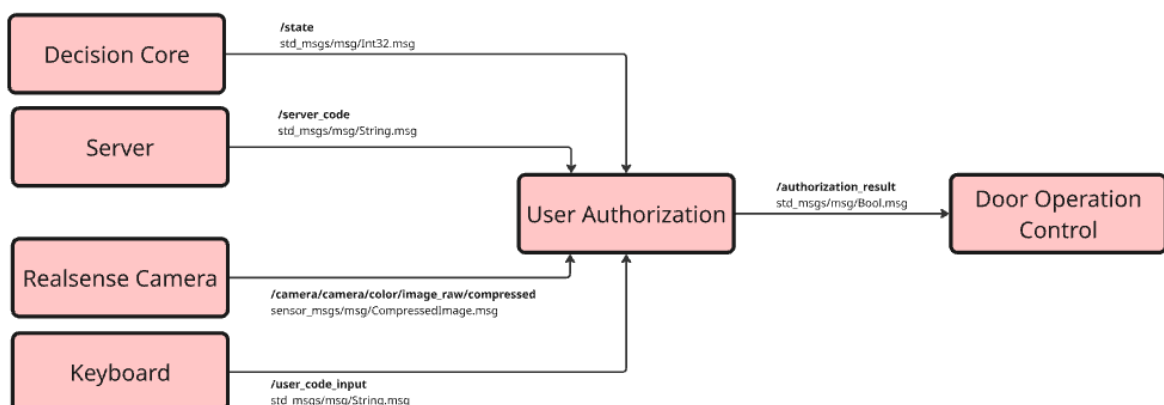
5. Architecture

a) Inputs:

1. QR code displayed in the user's smartwatch application and captured by the shuttle's RealSense camera
2. Unique code manually entered by the user via the keyboard
3. Authorization data associated with the user subscription, provided by the server
4. Shuttle state information, indicating that the shuttle is in the boarding state

b) Outputs:

1. Authorization result, which directly determines whether access to the shuttle is granted or denied and triggers the door-opening behaviour



6. Test Strategy

A. Test Basic

The system test cases for the feature **Secure User Authorization for Shuttle Access** are primarily derived from the refined feature requirements. Where applicable, relevant system-level requirements related to user authorization and boarding behaviour are referenced to ensure consistent behaviour across the complete user journey.

B. Test Techniques for Defining Test Cases

System testing for the feature **Secure User Authorization for Shuttle Access** is performed using black-box test design techniques, with scenario-based testing as the primary approach.

Scenario-based testing is selected to validate the feature from the user perspective under realistic operational conditions in the Model City environment. This approach allows verification of the complete user interaction flow during shuttle boarding.

The test design focuses on validating correct system behaviour for both successful and unsuccessful authorization attempts, as well as ensuring that authorization is only possible when the shuttle is in the appropriate boarding state. Negative and alternative flows are included to verify secure handling of invalid inputs and improper usage conditions.

C. Scenarios for Testing

Based on test analysis of the feature behaviour, user interaction flow, system states, and potential failure situations, the following system-level test scenarios are identified. These scenarios describe what needs to be tested at system level and serve as the basis for detailed test case design in the test specification.

Scenario 1: Successful QR Code-Based Authorization (Happy Flow)

This scenario verifies that a user can successfully authorize access to the shuttle by presenting a valid QR code when the shuttle is in the boarding state and all system components are functioning normally.

Scenario 2: QR Code-Based Authorization with Distance Variations from the RealSense Camera

This scenario verifies the system's behaviour when a valid QR code is presented at non-optimal distances from the RealSense camera during the boarding state, ensuring correct handling of distance-related recognition limitations.

Scenario 3: Successful Authorization via Manual Unique Code Entry

This scenario verifies that a user can successfully authorize access to the shuttle by entering a valid unique code using the keyboard when the shuttle is in the boarding state.

Scenario 4: Authorization Attempts with Invalid Credentials (Invalid QR Code, Invalid Unique Code)

This scenario verifies that the system correctly denies access when the user attempts authorization using invalid credentials, either through an invalid QR code or an invalid manually entered unique code.

Scenario 5: Authorization Attempt Outside the Boarding State (Shuttle Stopped at a Red Light)

This scenario verifies that user authorization is not permitted when the shuttle is not in the boarding state, even if valid authorization credentials are provided.

Scenario 6: User Authorization via Manual Code in Case of Camera Failure

This scenario verifies that the user can successfully authorize access using a valid unique code when the camera is unavailable, ensuring correct fallback behaviour during the boarding state.

Scenario 7: Authorization Failure Due to Camera Unavailability and Missing Unique Code

This scenario verifies that access to the shuttle is denied when no valid authorization method is available, such as when the camera is unavailable and the user does not provide a unique code.

Scenario 8: QR Code-Based Authorization Using Printed QR Code

This scenario verifies that the system can successfully authorize access when a valid QR code printed on paper is presented to the RealSense camera during the boarding state.

Scenario 9: QR Code-Based Authorization with Varying Display Light Intensity

This scenario verifies the system's robustness when a valid QR code is displayed under different screen brightness conditions, ensuring correct authorization behaviour during the boarding state.

D. Test Coverage

Test coverage for the feature **Secure User Authorization for Shuttle Access** is defined based on **requirements coverage** and **scenario coverage**.

a. **Requirements coverage:**

All refined feature requirements related to user authorization, access control, and boarding state behaviour are covered by at least one system test case. The target coverage is 100%, ensuring that every relevant feature requirement is verified during system testing.

b. **Scenario coverage:**

All identified test scenarios, including happy flows, variations, negative cases, and fallback situations, are covered by corresponding system test cases documented in the system test specification.

This coverage approach ensures that the feature is tested comprehensively under normal operating conditions as well as under exceptional and failure situations.

KPI: Authorization Decision Correctness

The system shall produce the correct authorization decision for all defined test scenarios. Successful authorization scenarios must result in access being granted, while invalid or forbidden scenarios must result in access being denied.

Target value: 100%.

E. Test Environment

System testing for the feature **Secure User Authorization for Shuttle Access** is conducted in the **Model City** test environment. This environment supports validation of user authorization behaviour under realistic operating conditions, including boarding state activation, user interaction, and system responses.

All defined test scenarios and corresponding system test cases are executed within this environment.